

Problem 7.3., 7.4: Determine the internal forces at point J and at point K of the structure shown.

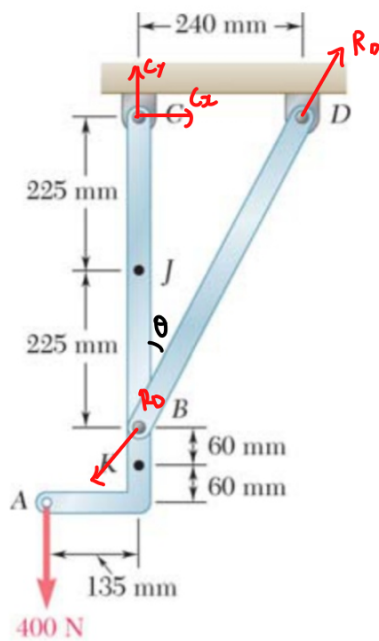


Fig. P7.3 and P7.4

$$\theta = \tan^{-1}(240/450) = 28.07^\circ$$

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$$\sum F_x = C_x + R_0 \sin \theta = 0$$

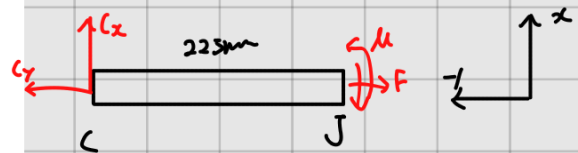
$$\sum F_y = C_y + R_0 \cos \theta - 400 = 0$$

$$\sum M_C = +450 \sin \theta \times R_0 + 135 \times 400 = 0$$

$$R_0 = -255.0 \text{ N}$$

$$C_x = 119.9 \text{ N}$$

$$C_y = 625.0 \text{ N}$$



$$\sum M_J = 0$$

$$\sum F_x = C_x - V = 0$$

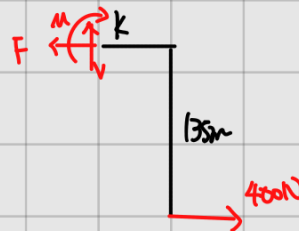
$$\sum F_y = C_y - F = 0$$

$$\sum M_J = -225 \times C_x + M = 0$$

$$\therefore V = 119.9 \text{ N}$$

$$F = 625.0 \text{ N}$$

$$M = 27.00 \text{ N}\cdot\text{m}$$



$$\sum F_x = 0$$

$$\sum F_x = -F + 400 = 0$$

$$\sum F_y = V = 0$$

$$\sum M_K = -M + 135 \times 400 = 0$$

$$\therefore V = 0$$

$$F = 400$$

$$M = 54 \text{ N}\cdot\text{m}$$

Problem 7.7, 7.8: Two members, each consisting of a straight and a quarter-circular portion of rod, are connected as shown and support a 75-N load at A. Determine the internal forces at point J and at point K.

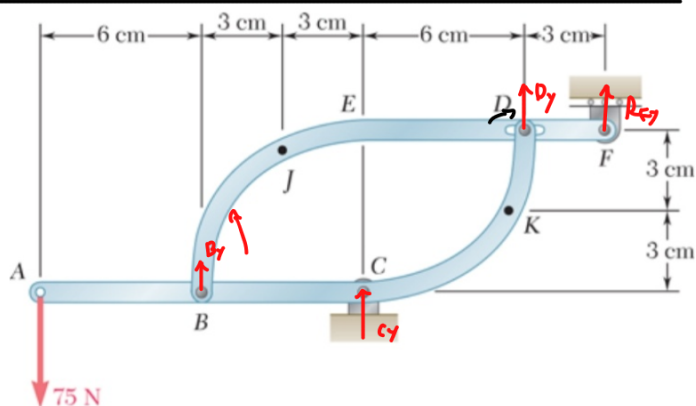
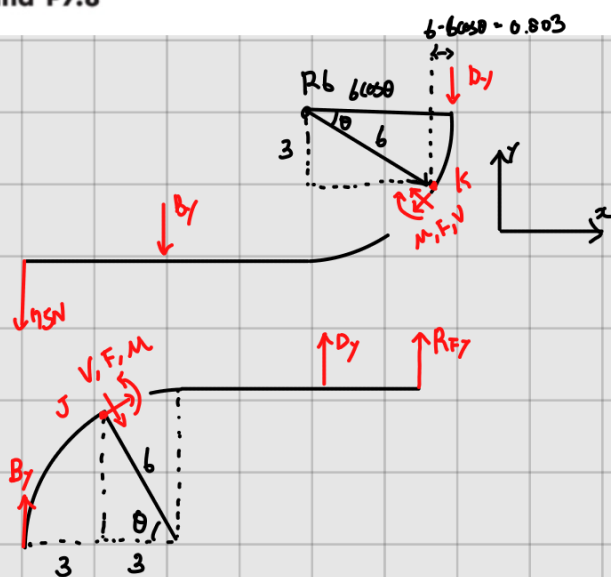


Fig. P7.7 and P7.8



전체로써

$$\sum F_x = C_x = 0$$

$$\sum F_y = R_{Ey} + C_y - 75 = 0$$

$$\sum M_c = 7 \times R_{Ey} + 12 \times 75 = 0$$

$$R_{Ey} = -100 \text{ N}$$

$$C_y = 175 \text{ N}$$

$$C_x = 0$$

ACD member

$$\sum F_x = -B_x + C_x = 0$$

$$\sum F_y = -75 - B_y + C_y - D_y = 0$$

$$\sum M_B = 6 \times 75 + 6 \times C_y - 12 D_y = 0$$

$$D_y = 125 \text{ N}$$

$$B_y = -25 \text{ N}$$

Point K: $\sin \theta = \frac{1}{2}$, $\cos \theta = \frac{\sqrt{3}}{2}$

$$\sum F_x = -F \sin \theta - V \cos \theta = 0$$

$$\sum F_y = -F \cos \theta + V \sin \theta - D_y = 0$$

$$\sum M_K = -M - 0.803 \times D_y = 0$$

$$\begin{cases} \therefore M = -1.003 \text{ N}\cdot\text{m} \\ F = -108.25 \text{ N (compression)} \\ V = 62.5 \text{ N} \end{cases}$$

Point J: $\cos \theta = \frac{1}{2}$, $\sin \theta = \frac{\sqrt{3}}{2}$

$$\sum F_x = F \sin \theta + V \cos \theta = 0$$

$$\sum F_y = F \cos \theta - V \sin \theta + B_y = 0$$

$$\sum M_J = M - 3 B_y = 0$$

$$\therefore M = 3 B_y = -0.75 \text{ N}\cdot\text{m}$$

$$F = 12.5 \text{ N (tension)}$$

$$V = -21.6 \text{ N}$$

Problems 7.29, 7.30, 7.32, 7.35, 7.40, 7.47, 7.49, 7.51:
For the beam and loading shown, (a) draw the SFD and BMD, (b) determine the maximum absolute values of the shear and bending moment.

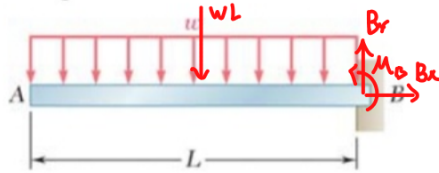


Fig. P7.29

7.29.

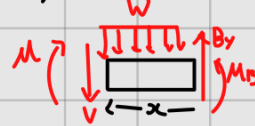
$$\sum F_x = B_x = 0$$

$$\sum F_y = -WL + B_y = 0$$

$$\sum M_B = M_B + \frac{1}{2} \times WL = 0$$

$$M_B = -\frac{WL^2}{2}$$

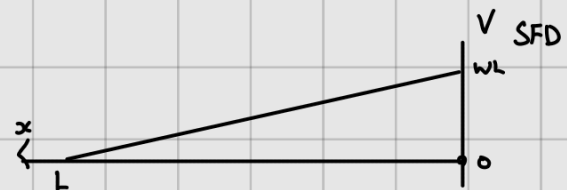
$$B_y = WL$$



$$\sum F_y = -V - wx + B_y = 0$$

$$\sum M_x = -M + M_B - \frac{x}{2} \times wx + x \times B_y = 0$$

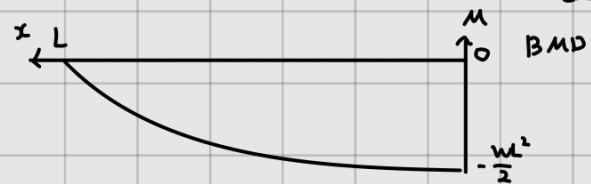
$$V = wx - B_y = -w(L - x)$$



$$M = M_B + B_y x - \frac{w}{2} x^2$$

$$= -\frac{w}{2} x^2 + WLx - \frac{WL^2}{2}$$

$$\frac{dM}{dx} = V \quad (\text{ok})$$



$$b) \begin{cases} V_{\max} = WL \\ M_{\min} = -\frac{WL^2}{2} \end{cases}$$

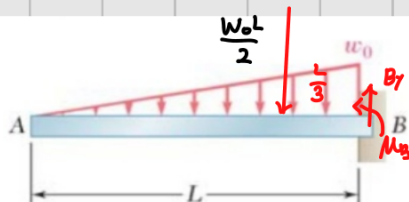


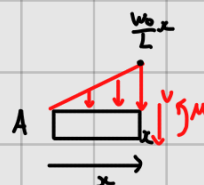
Fig. P7.30

$$\sum F_x = B_x = 0$$

$$\sum F_y = B_y - \frac{w_0 L}{2} = 0$$

$$\sum M_B = M_B + \frac{1}{3} \times \frac{w_0 L}{2} = 0$$

$$B_y = \frac{w_0 L}{2}, \quad M_B = -\frac{w_0 L^2}{6}$$

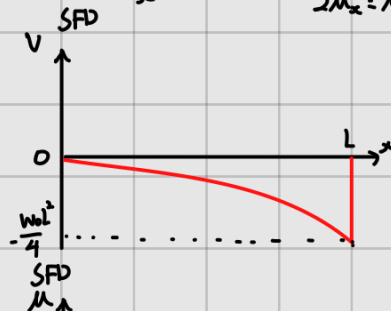


$$\sum F_y = -\left(\frac{w_0 x}{2} \times x \times \frac{1}{2}\right) - V = 0$$

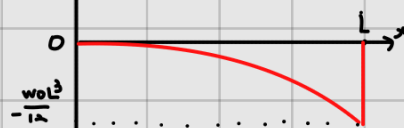
$$V = -\frac{w_0 x^2}{4}$$

$$\sum M_x = M + \frac{x}{3} \times \frac{w_0 x^2}{4} = 0$$

$$M = -\frac{w_0 x^3}{12} \quad \frac{dM}{dx} = V \quad (\text{ok})$$



$$V_{\min} = -\frac{w_0 L^2}{4}$$



$$M_{\min} = -\frac{w_0 L^3}{12}$$

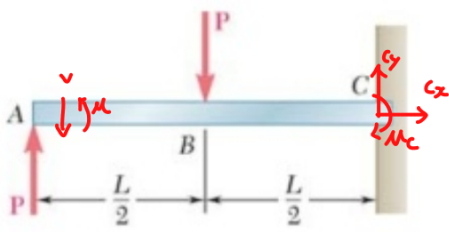


Fig. P7.32

7.32

$$\sum F_y = P - P + C_y = 0$$

$$\sum F_x = C_x = 0$$

$$\sum M_B = -PL + \frac{PL}{2} - M_c = 0$$

$$C_x = C_y = 0$$

$$M_c = \frac{PL}{2}$$

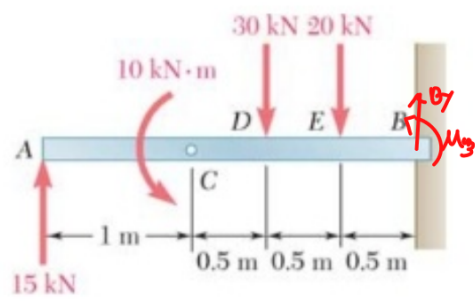
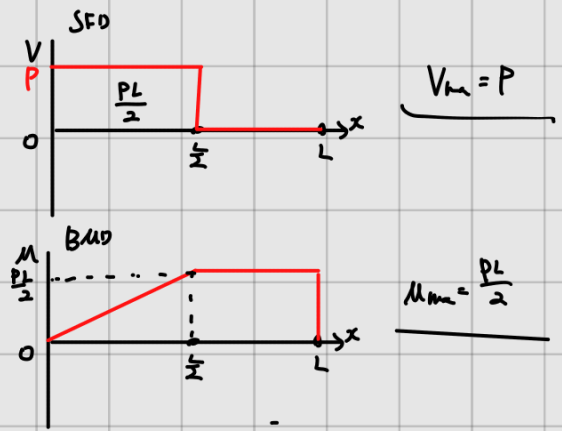


Fig. P7.35

7.35

$$\sum F_y = 15 - 30 - 20 + B_y = 0$$

$$B_y = 35 \text{ kN}$$

$$\sum M_B = -2.5 \times 15 + 10 + 1 \times 30 + 0.5 \times 20 + M_B = 0$$

$$M_B = -12.5 \text{ kN}\cdot\text{m}$$

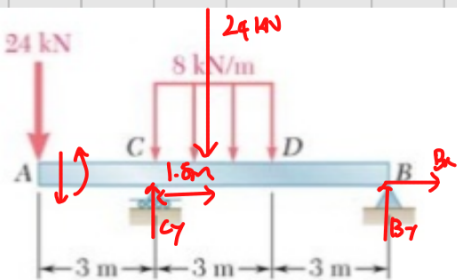
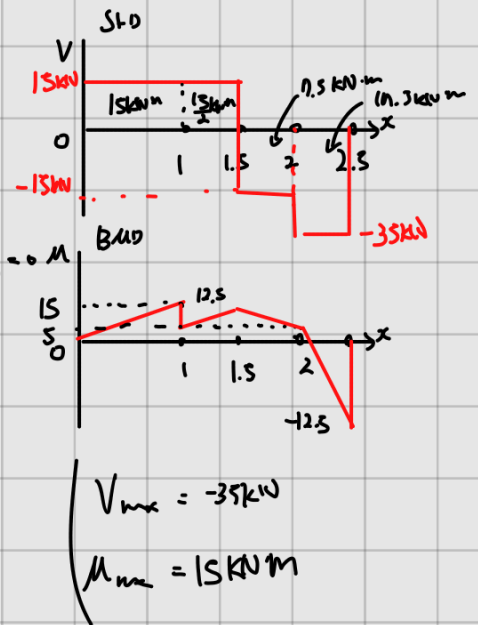


Fig. P7.40

7.40

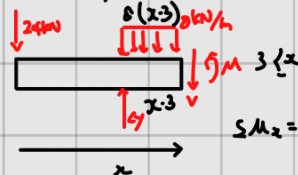
$$\sum F_x = B_x = 0$$

$$\sum F_y = -24 + C_y - 24 + B_y = 0$$

$$\sum M_B = 4.5 \times 24 - 6 \times C_y + 9 \times 24 = 0$$

$$\therefore C_y = 54 \text{ kN}$$

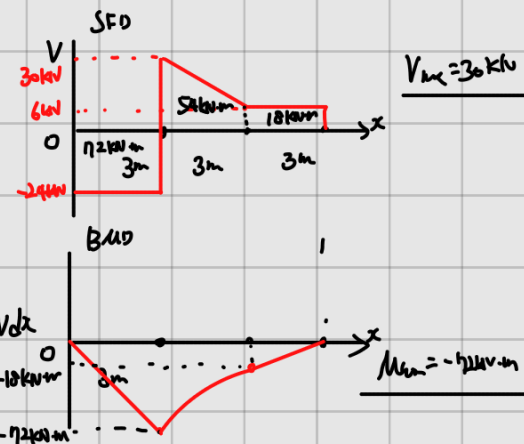
$$B_y = -6 \text{ kN}$$



$$\sum M_x = M + \frac{x-3}{2} \times 8(x-3) - (x-3) \times 54 + x \times 24 = 0$$

$$M = -4(x-3)^2 + 54(x-3) - 24x$$

$$= -4x^2 + 54x - 198$$



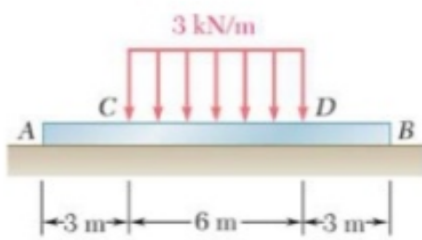
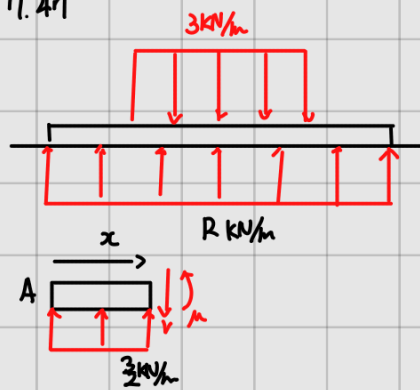


Fig. P7.47

7.47



ΣFy = 0

$$\Sigma F_y = -3 \text{ kN/m} \times 6 \text{ m} + R \text{ kN/m} \times 12 \text{ m} = 0$$

$$R = \frac{3}{2}$$

$0 \leq x < 3 \text{ m}$

$$\Sigma F_y = \frac{3}{2}x - V = 0$$

$$V = \frac{3}{2}x$$

$$\Sigma M_x = M - \frac{3}{2}x \times \frac{x}{2} = 0$$

$$M = \frac{3}{4}x^2$$

$3 \leq x < 9 \text{ m}$

$$\Sigma F_y = \frac{3}{2} \times 3 - \frac{3}{2} \times (x-3) - V = 0$$

$$V = -\frac{3}{2}x + 9$$

$$\Sigma M_x = \frac{3}{2} \times (x-3) \times \frac{(x-3)}{2} - (x-\frac{3}{2}) \times \frac{9}{2} + M = 0$$

$$M = -\frac{3}{4}x^2 + 9x - \frac{27}{2}$$

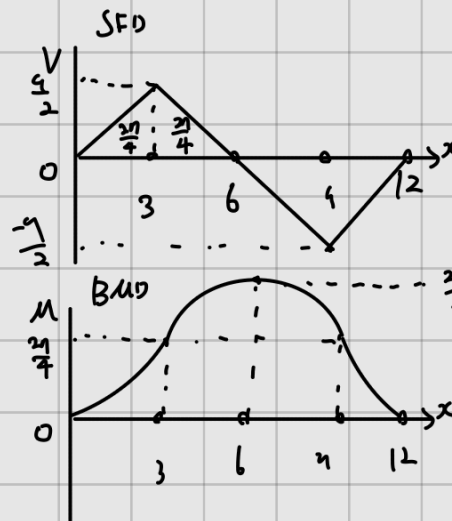
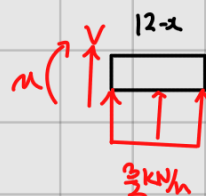
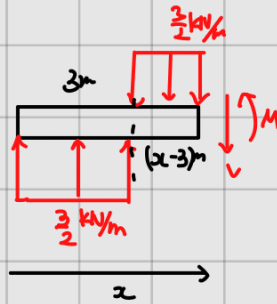
$9 \leq x < 12 \text{ m}$

$$\Sigma F_y = \frac{3}{2}(12-x) + V = 0$$

$$V = \frac{3}{2}x - 18$$

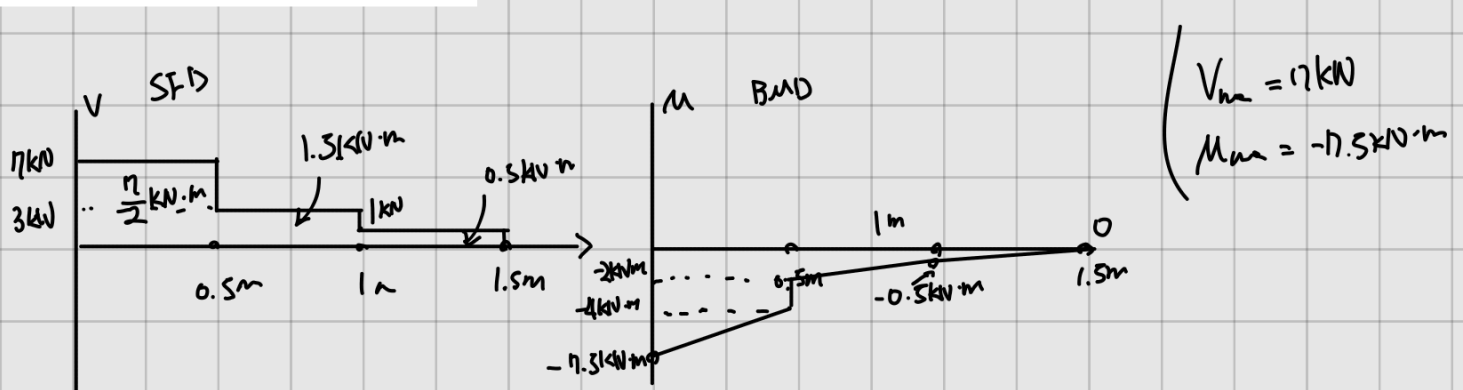
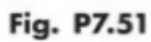
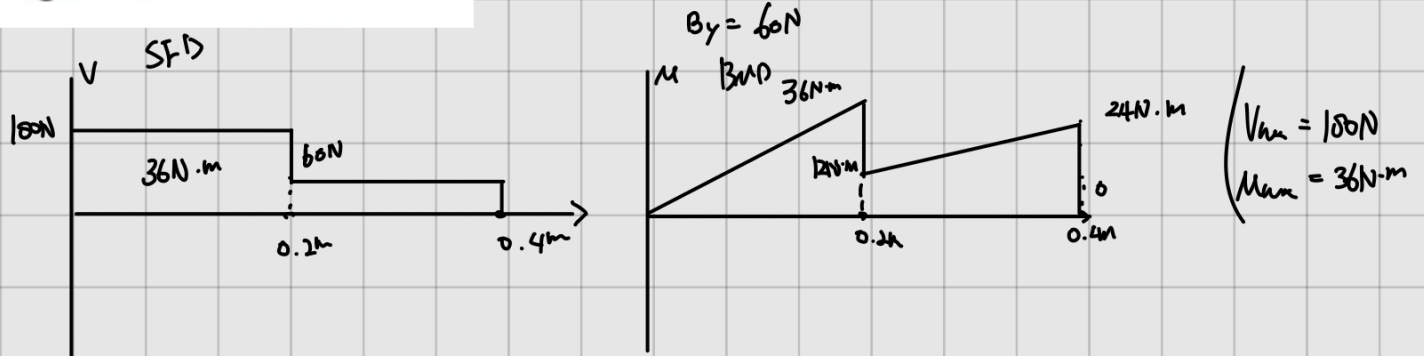
$$\Sigma M_x = -M + \frac{(12-x)}{2} \times \frac{3}{2}(12-x) = 0$$

$$M = \frac{3}{4}(12-x)^2$$



$$V_{max} = \frac{27}{4} \text{ kN}$$

$$M_{max} = \frac{27}{4} \text{ kN-m}$$



Problems 7.75, 7.76, 7.77, 7.83, 7.87, 7.88: For the beam and loading shown, (a) draw the SFD and BMD, (b) determine the maximum absolute values of the shear and bending moment using the method of Sec. 7.6.

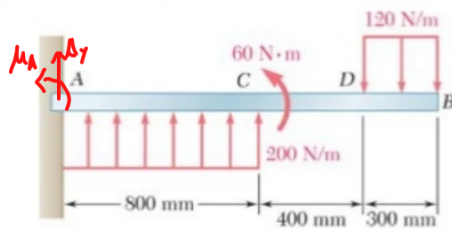


Fig. P7.75

7.75.

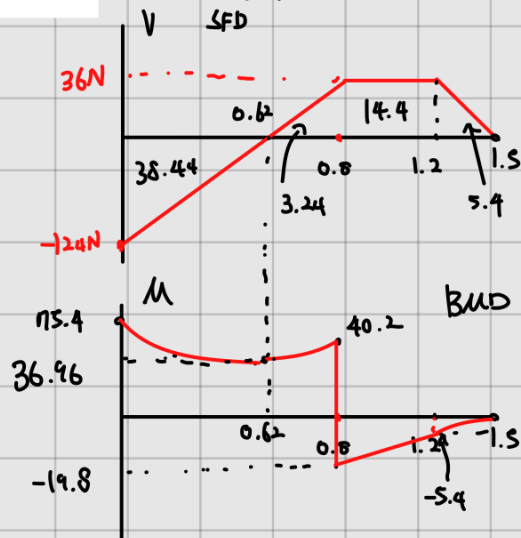
$$\sum F_x = A_x = 0$$

$$\sum F_y = A_y + 0.8 \times 200 - 0.3 \times 120 = 0$$

$$\sum M_A = M_A + 0.4 \times 160 + 60 - 1.35 \times 36 = 0$$

$$A_y = -124 \text{ N}$$

$$M_A = -75.4 \text{ N}\cdot\text{m}$$



$$\frac{dV}{dx} = -w$$

$$\frac{dM}{dx} = V$$

$$\begin{cases} V_{max} = -124 \text{ N} \\ M_{max} = 75.4 \text{ N}\cdot\text{m} \end{cases}$$

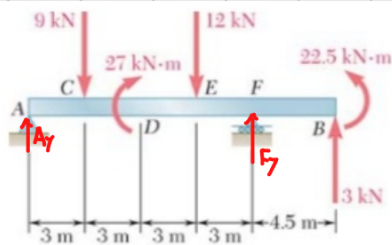


Fig. P7.76

7.76.

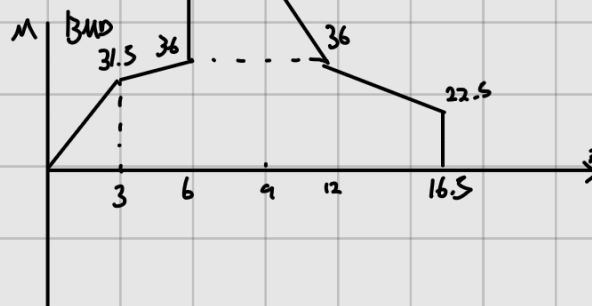
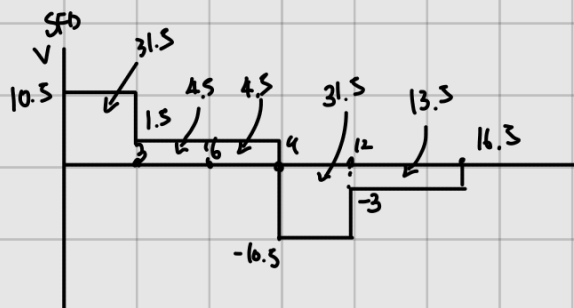
$$\sum F_x = A_x = 0$$

$$\sum F_y = A_y - 9 - 12 + F_y + 3 = 0$$

$$\sum M_A = -3 \times 9 - 27 - 9 \times 12 + 12 \times F_y + 16.5 \times 3 + 22.5 = 0$$

$$F_y = 17.5 \text{ kN}$$

$$A_y = 10.5 \text{ kN}$$



$$\begin{cases} V_{max} = 10.5, -16.5 \text{ kN} \\ M_{max} = 63 \text{ kN}\cdot\text{m} \end{cases}$$

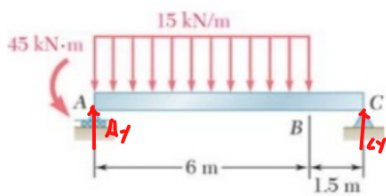
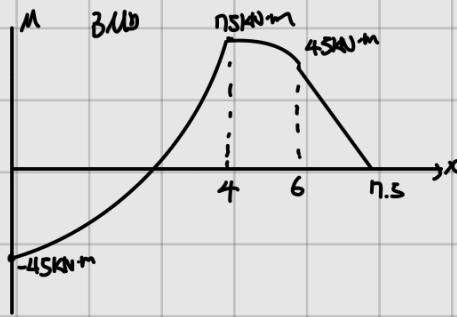
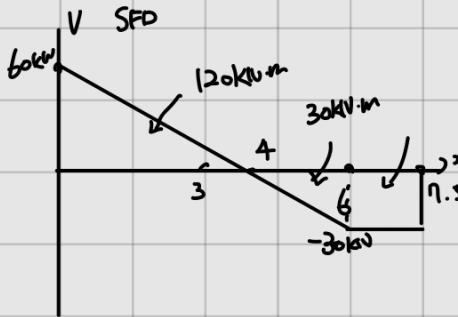


Fig. P7.77



7. m
 $\sum F_x = C_x = 0$
 $\sum F_y = A_y - 15 \times 6 + C_y = 0$
 $\sum M_A = 45 - 3 \times 90 + 7.5 \times C_y = 0$
 $C_y = 30 \text{ kN}$
 $A_y = 60 \text{ kN}$

$V_{max} = 60 \text{ kN}$
 $M_{max} = 75 \text{ kN}\cdot\text{m}$

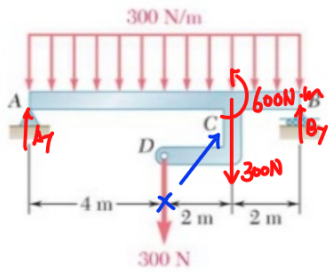
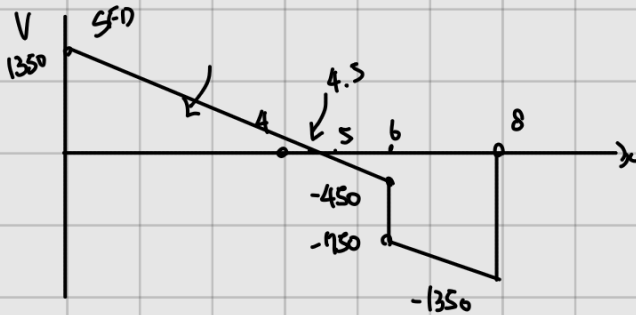
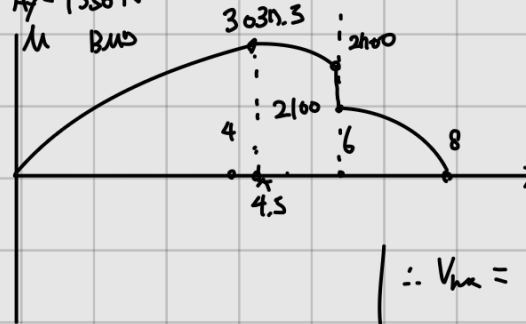


Fig. P7.83



7.83
 $\sum F_x = A_x = 0$
 $\sum F_y = A_y + B_y - 300 - 300 \times 8 = 0$
 $\sum M_A = -6m \times 300 \text{ N} + 8m \times B_y - 4 \times 2400 + 600 = 0$
 $B_y = 1350 \text{ N}$
 $A_y = 1350 \text{ N}$



$\therefore V_{max} = 1350, -1350 \text{ N}$
 $M_{max} = 3037.5 \text{ N}\cdot\text{m}$

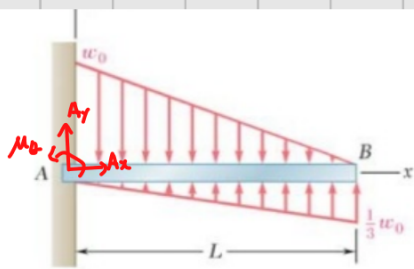


Fig. P7.87

$$\begin{aligned}\sum F_x &= A_x = 0 \\ \sum F_y &= A_y - \frac{w_0 L}{2} + \frac{1}{6} w_0 L = 0 \\ \sum M_A &= M_A - \frac{1}{3} \times \frac{w_0 L}{2} + \frac{1}{6} w_0 L \times \frac{2}{3} L = 0 \\ A_y &= \frac{1}{3} w_0 L \\ M_A &= \frac{1}{10} w_0 L^2\end{aligned}$$

$$\begin{aligned}V(0) &= \frac{1}{3} w_0 L \\ M(0) &= -\frac{1}{10} w_0 L^2\end{aligned}$$

$$\begin{aligned}P_1(0) &= -w_0 \\ P_1(L) &= 0 \\ W_1 &= \frac{w_0}{L} (x-L)\end{aligned}$$

$$\begin{aligned}P_2(0) &= 0 \\ P_2(L) &= \frac{1}{3} w_0 \\ W_2 &= \frac{w_0}{3L} x\end{aligned}$$

$$W(x) = W_1 + W_2 = \frac{4w_0}{3L} x - w_0$$

$$\frac{dV}{dx} = W$$

$$V(x) = \frac{2w_0}{3L} x^2 - w_0 x + C_1$$

$$V(0) = \frac{1}{3} w_0 L \quad \therefore C_1 = \frac{1}{3} w_0 L$$

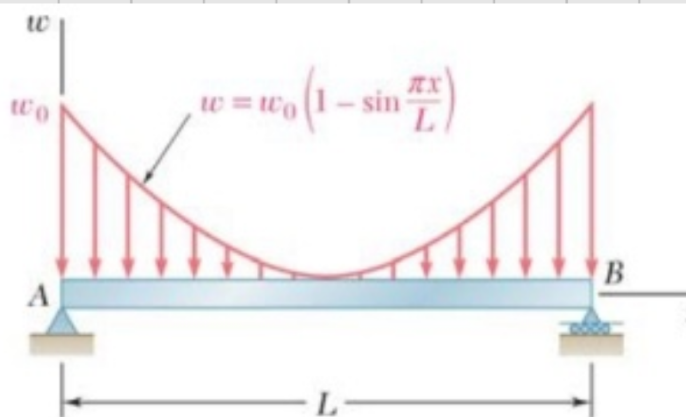
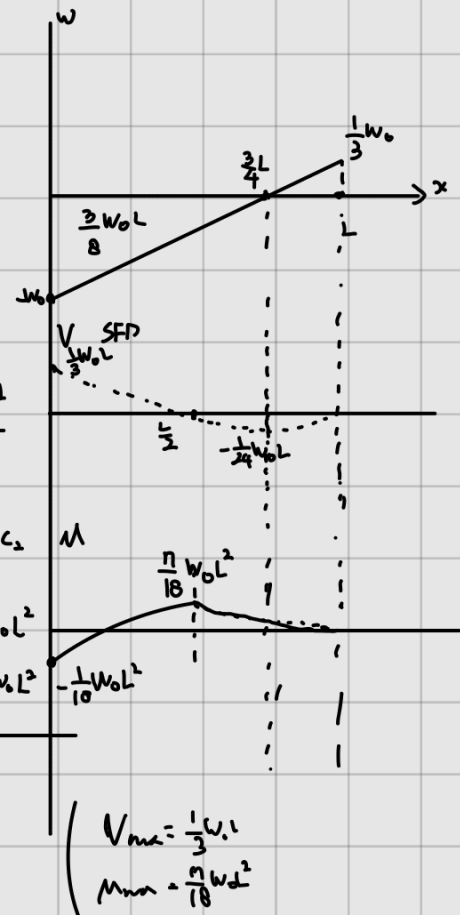
$$V(x) = \frac{2w_0}{3L} x^2 - w_0 x + \frac{1}{3} w_0 L$$

$$\frac{dM}{dx} = V$$

$$M(x) = \frac{2w_0}{9L} x^3 - \frac{w_0}{2} x^2 + \frac{1}{3} w_0 L x + C_2$$

$$M(0) = -\frac{1}{10} w_0 L^2 \quad \therefore C_2 = -\frac{1}{10} w_0 L^2$$

$$M(x) = \frac{2w_0}{9L} x^3 - \frac{w_0}{2} x^2 + \frac{1}{3} w_0 L x - \frac{1}{10} w_0 L^2$$



$$W = w_0 \left(1 - \sin \frac{\pi x}{L}\right)$$

$$V = \int w_0 \sin \frac{\pi x}{L} - w_0 dx = -\frac{w_0 L}{\pi} \cos \frac{\pi x}{L} - w_0 x + C_1$$

$$\text{At } x = \frac{L}{2}, V = 0 \quad \therefore C_1 = \frac{w_0 L}{2}$$

$$V(x) = -\frac{w_0 L}{\pi} \cos \frac{\pi x}{L} - w_0 x + \frac{w_0 L}{2}$$

$$M(x) = \int V(x) dx = \frac{-w_0 L^2}{\pi^2} \sin \frac{\pi x}{L} - \frac{w_0}{2} x^2 + \frac{w_0 L}{2} x + C_2$$

$$M(0) = 0 \quad \therefore C_2 = 0$$

$$M(x) = -\frac{w_0 L^2}{\pi^2} \sin \frac{\pi x}{L} - \frac{w_0}{2} x^2 + \frac{w_0 L}{2} x$$

